Corollary

Let n be a positive integer. Then

$$\sum_{k=0}^{n}(-1)^k\binom{n}{k}=0$$

- **Proof**

Using the binomial theorem with $x=1$ and $y=-1$, we can get

$$0=(1+(-1))^n=\sum_{k=0}^{n}\binom{n}{k}1^{n-k}(-1)^k=\sum_{k=0}^{n}\binom{n}{k}(-1)^k$$

This proves the corollary.